# P3CTeX: reusable student tools

P3CTeX

Development guide, October 2026

## 1 Choose the tools you need

```
\documentclass[code,images,tables,diagrams,
               calculations,requirements]{P3CTeX}
```

Existing module options remain supported. The diagram entry point `pxDIAG` uses the established `pxCMP` engine and presets. The new calculation package is `pxCALC`. Package names do not dictate the keywords used to load tools.

## 2 Write a formula once

```
\pxDefVar{m}{1200}[m]
\pxDefVar{v}{25}[v]
\pxDefFormula{Ek}{\frac{1}{2}[m][v]^2}
  [symbol=E_k,unit=J]
\pxSolve{Ek}
\pxSolve[values={m=800,v=40}]{Ek}
```

$$E_k = \frac{1}{2}mv^2 = \frac{1}{2}(1200)(25)^2 = 375\,000\,\mathrm{J}$$

$$E_k = \frac{1}{2}mv^2 = \frac{1}{2}(800)(40)^2 = 640\,000\,\mathrm{J}$$

The override is local. The next call still uses the original values. `steps=1` prints the result; `steps=2` also prints the formula; `steps=3` also prints substituted values. Settings can be supplied with `\pxCalcSetup`, with a formula, or for one call.

`\pxCalcStore{Ek}{\energy}` saves the unrounded number for reuse. `\pxSetVar{v}{30}` changes the value in the current TeX group. `\pxCalc{sin(pi/2)}` evaluates a numerical expression directly. Units format results; they do not convert values. Divide joules by 1000 before labelling the result as kilojoules.

## 3 Verbatim terminal and source ranges

Shell session

```
$ printf '%s\n' "$HOME" '100%'
/home/student
100%
```

```
\begin{pxTerminal}{Shell session}
$ printf '%s\n' "$HOME" '100%'
/home/student
100%
\end{pxTerminal}
\pxCodeIn[pxBash]|echo "$HOME" # literal code|
\pxCodeChunk[pxJava]{Main.java}{12}{20}{Validation}
```

Terminal options use listings keys, like `pxCode`. Palettes and language definitions remain customisable through the existing listing API. Inline verbatim must be outside other commands' arguments.

# 4 Requirements and analysis

**User story: US-102: Reuse calculations**

**As a:** student
**I want:** to change inputs without copying the formula
**So that:** the displayed working stays consistent
**Acceptance criteria:** Show substitutions and preserve the defaults.

**Scenario: Change one input**

**Given:** a speed of 25 metres per second
**When:** I solve with a speed of 30
**Then:** the result is 540 000 joules

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| **Strengths** | **Weaknesses** |
|---|---|
| Reusable formulas | Explicit supported syntax |
| **Opportunities** | **Threats** |
| Student laboratory reports | Incorrect input data |

Cards accept normal `tcolorbox` options; their labels are editable with `\pxREQsetup{labels={as-a=Com a,i-want=Vull`
The SWOT grid reuses `pxTAB`; its optional argument accepts table options, including an explicit column specification.

# 5 Reuse the existing environment

Safe images, placeholders and image lists already belong to `pxSRC`. Transition tables already include partition labels in `pxCMP`. Use `pxMEM` for report annexes and the class for cover metadata and authorship declarations. These tools do not introduce replacement systems.

The companion `student-tools.md` gives the complete syntax limits, customisation keys and migration details.